



Berkshire Alpha

An autonomous options agent
whose models argue — and
whose risk layer cannot be
argued with.

4 models · 3 vendors · 9 routed nodes

Trades a language model can authorise: 0

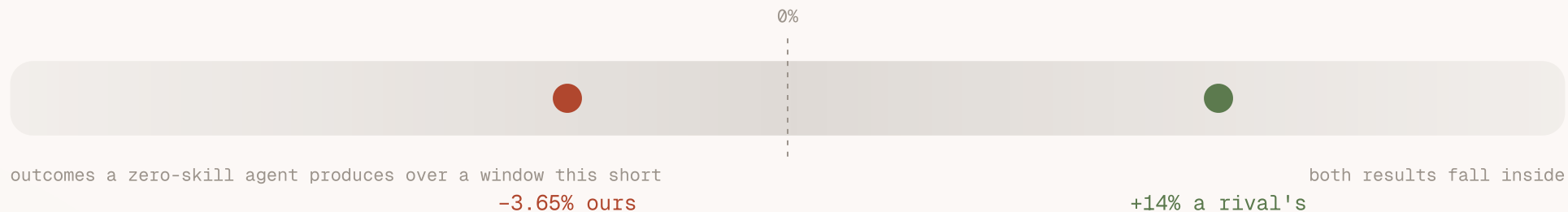
557 tests · 267 commits

berkshire-alpha.vercel.app

3-7 DTE vertical spreads · Python/FastAPI + Next.js
Postgres on Railway · CI on every push · MIT

Over four sessions, P&L cannot tell skill from luck.

Four sessions. Around ten trades. That window is noise, and every quant in this room already knows it. We compute how much: Deflated Sharpe and Minimum Track Record Length, against a counted trial budget of $N = 16$.



MinTRL says a record this short cannot demonstrate skill at any Sharpe we could plausibly have. That is true of our -3.65% , and equally true of somebody else's $+14\%$. So judge the architecture.

Every AI trading agent has an opinion. Almost none of them have a constraint.

	GATE: APPROVE	GATE: REFUSE
LLM: APPROVE	REACHABLE ORDER BUILT & SENT gate sized it at quarter-Kelly · 7 fills on the tape	REACHABLE · HAPPENED NO_TRADE ids 33 · 36 · 136 · 168 – two of them after a 3-0 unanimous approve
LLM: REFUSE	UNREACHABLE NOTHING TO APPROVE the gate reads proposals; it never manufactures one	REACHABLE · HAPPENED NO_TRADE CONSERVATIVE dissented on 7 of 12 votes

The category's failure mode is a language model holding order-placement authority. Here the models produce evidence and a deterministic Python gate produces the decision.

Enforced structurally, not by convention

`test_gate_never_sees_llm`
`test_confidence_never_reaches_sizing_or_gates`
`test_agents_never_execute`

The number of trades a language model can authorise in this system is zero — and three tests fail the build if that ever changes.

Design choices with a test behind each one.

“When two models that share no weights agree, that's evidence. When one model wears two hats and agrees with itself, that's an artefact.”

Disagree-or-Commit

Agreement counts only as a commit citing new evidence, against a 0.85 bar. Never met.

Scored $0.70 \times \text{commit} + 0.30 \times \text{grounding}$. Bull/Bear routed to different families in config; per-node routing shipped 2 Sep, after the last debated cycle – replayable transcripts show Qwen2.5-72B on every node.

Macro, quarantined

It can change what we look at. It can never change how much we risk.

GLD / USO / IBIT on two horizons → one dataclass of selection parameters.
`test_macro_tuning_fields_are_selection_only` fails the build if a sizing field is added.

Kelly that can only reduce

$f^* \times 0.25$ under a hard 2% cap — cut from half after two losing closes.

KELLY_FRACTION $0.50 \rightarrow 0.25$ the night of 1 Sep, 0 wins in 2 closed trades, reasoning in the commit message.

The unanimous approval the gate refused.

QUANT	NVDA · spot 217.865 · IV_atm 0.35645 · RV_20 0.4550 · VRP 0.7834	REGIME DEBIT
DEBATE	2 rounds · consensus 0.65 against a 0.85 bar · conviction 0.50	UNRESOLVED
TRADER	BUY 215 C / SELL 220 C · exp 2026-09-04 · confidence 0.65	BULL_CALL_SPREAD
RISK VOTES	AGGRESSIVE APPROVE · NEUTRAL APPROVE · CONSERVATIVE APPROVE	UNANIMOUS 3-0
GATE	NEGATIVE_EDGE · observed -0.00448 against threshold 0.0	NO_TRADE

CONSERVATIVE persona note, verbatim

“The max loss per spread is within acceptable limits and the risk/reward ratio is favorable.”

Kelly f* came out below zero, and a sizing function that can only reduce has nowhere to go from there.

curl .../decisions/36 · re-pullable live

Three personas approved it. The arithmetic said the edge was negative, and the arithmetic is the one that signs.

Our LLM wanted to trade. Our gate said no. For two days.

Our own Reflector examined the freeze and voted to keep it. The machine argued itself into agreeing with the cage.



31 Aug · CLI could not verify account → HALT · 78 samples · range \$0.00

2-3 Sep · portfolio delta breach → ENTRIES FROZEN · reduce_only = true

Endpoints from GET /equity/history · intraday path between them drawn schematically

DELTA VS MANDATE

222%

-\$32,125.96 vs ±\$14,453.10

EVALUATED

350

action=NO_TRADE, 350/350

REFUSED ON REDUCE_ONLY

46

18 spreads built

INFERENCE SPENT

\$0.00

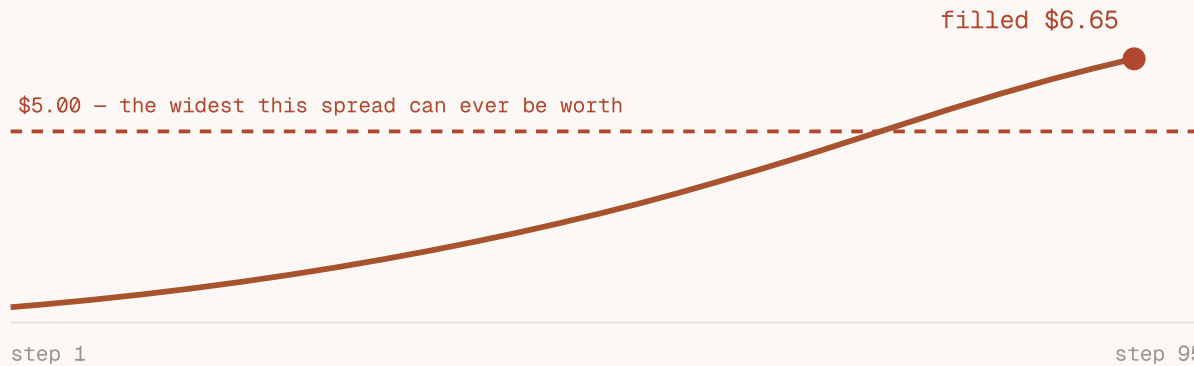
0 debates · 0 trades

REFLECTOR · /REFLECTIONS ID 3

“the session resulted in 0 trades entered, indicating the agent's discipline in adhering to risk controls.”

THE DEFECT, AND WHAT IT COST

−3.65%. One fill. Root-caused, reproduced, fixed mid-competition.



95 replace events, LLY · walk telemetry, reproduced under test

SLIPPAGE ON ONE FILL

\$1,884

of \$1,961 total – 96%

The walk cap was purely relative: $\text{mid} + 0.70 \times (\text{natural} - \text{mid})$. On a chain quoting 8.90 / 15.09 — 52% wide — natural was 4.6× mid.

5 of 7 fills landed at exactly theoretical mid: the paper-fill illusion, quantified in our own submission.

−3.65%, and we can name every basis point of it. Judge us on whether the architecture caught it. It did, in 300 seconds.

The CLI isn't a checkbox. It's the source of truth, and it can halt us.

```
GET /tools/usage · live counters
```

tool calls, lifetime	736	failures: 0
cli · get_account	290 ms	x266
cli · list_positions	282 ms	x266
option chains	55 ms/sym	30 in 1.6 s
rate budget used	26%	52 req/cycle of 200/min

It can stop the agent

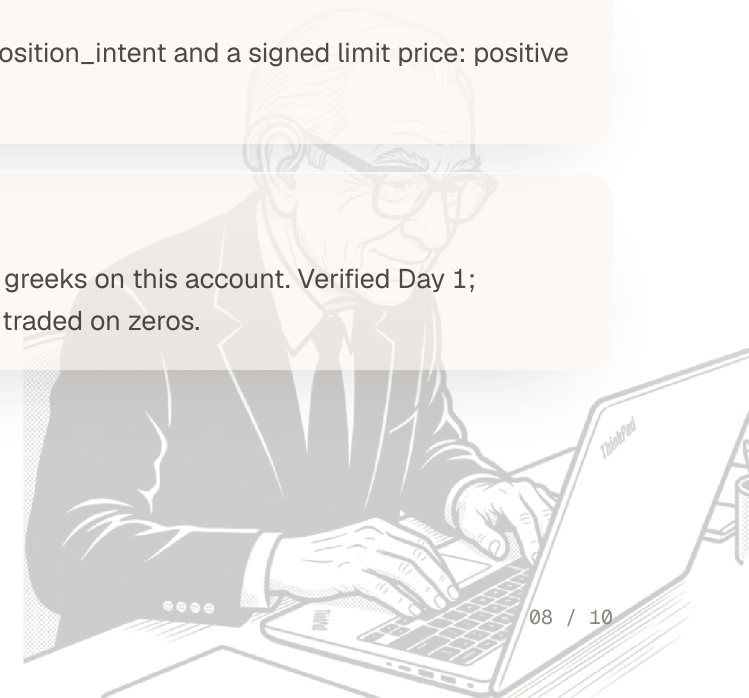
cli_bridge.py is what the Fund Manager gate reads for account state. If it cannot reach the account, the agent halts rather than trade on unverified state.

Multi-leg, sign-disciplined

alpaca-py submits mleg orders with position_intent and a signed limit price: positive is a net debit, negative a net credit.

Greeks from indicative

The default opira feed returns all-zero greeks on this account. Verified Day 1; degenerate blocks are dropped, never traded on zeros.



Four things we found before you did.

Sealed window · criterion 1 (positive realised P&L): NOT MET · criterion 2 (no gate hand-tuned in response to what it rejected): MET

The macro ladder's risk-off rung is inverted

RISK_OFF sets a looser momentum bar than NEUTRAL. Found inside the sealed window — an open item, not a commit.

`pre-registration forbade touching it mid-window`

Paper fills are not real fills

5 of 7 filled at exactly theoretical mid — quantified, not banked.

`docs/friction.md`

Our backtest harness had a VRP tautology

Every replay came out credit-only by construction.

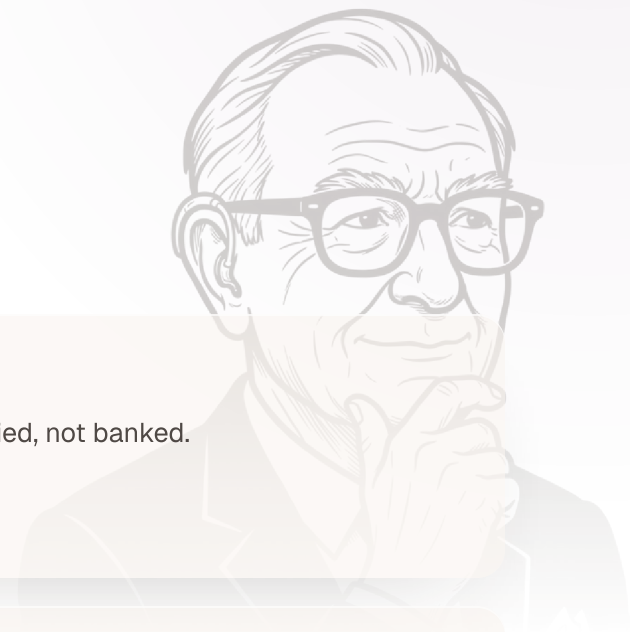
`audited in docs/report.md — not discovered by a judge`

Our ledger disagreed with the broker by \$224

Found it, published it, did not backfill.

`reconciliation · unresolved, stated`

None of these is a rough edge we ran out of time to sand. Each is a place where the honest answer was to say it out loud.



\$0.0033 per fully-deliberated candidate. This scales.

\$0.0394

total inference, whole competition
183 calls · 146,210 tokens · against a \$16 budget = 0.25% utilisation

calls = 2S + 7.3D

predicted 27.9 · measured 28 · universe size does not appear in it, so widening from 50 names to 500 costs zero additional tokens

PER SCAN CYCLE

\$0.0058

28 calls · 21,851 tokens · 67 s

PER ORDER SENT

\$0.0049

7 fills on the tape

REGULATORY FRICTION

\$5.21

whole competition

TOOL CALLS

736 / 0 failed

GET /tools/usage

We built an agent that knows how to do nothing. It is the only capability in this category that is actually hard.